

14.5 Solution: (a) For the particle with energy E in a repulsive potential, the shortest distance it can get to the central field with head-on collision is the solution to $E = V(r_{\min})$. From Lamour's formula, the instantaneous power radiated is

$$Pdt = \frac{2}{3} \frac{z^2 e^2}{c^3} |\dot{\mathbf{v}}|^2 dt.$$

For a nonrelativistic particle, using Newton's second law, we know

$$m\dot{\mathbf{v}} = -\nabla V = -\frac{\partial V}{\partial r} \mathbf{e}_r,$$

for head-on collision. Also, from energy conservation,

$$E = V(r_{\min}) = \frac{1}{2}mv^2 + V(r),$$

we can express the velocity as

$$v = \sqrt{\frac{2}{m} [V(r_{\min}) - V(r)]}^{1/2}.$$

Then, we can write the instantaneous power radiated as

$$Pdt = \frac{2}{3} \frac{z^2 e^2}{c^3} |\dot{\mathbf{v}}|^2 \frac{dr}{v} = \frac{2}{3} \frac{z^2 e^2}{m^2 c^3} \sqrt{\frac{m}{2}} \left| \frac{\partial V}{\partial r} \right|^2 \frac{dr}{\sqrt{V(r_{\min}) - V(r)}}.$$

The total energy radiated, taking into account that the particle will be reflected, now becomes

$$\Delta W = \frac{4}{3} \frac{z^2 e^2}{m^2 c^3} \sqrt{\frac{m}{2}} \int_{r_{\min}}^{+\infty} \left| \frac{\partial V}{\partial r} \right|^2 \frac{dr}{\sqrt{V(r_{\min}) - V(r)}}.$$

(b) For Coulomb potential,

$$V(r) = \frac{zZe^2}{r},$$

the integral becomes

$$\begin{aligned} \Delta W &= \frac{4}{3} \frac{z^2 e^2}{m^2 c^3} \sqrt{\frac{m}{2}} (zZe^2)^{3/2} \int_{r_{\min}}^{+\infty} \frac{1}{r^4} \frac{dr}{\sqrt{r_{\min}^{-1} - r^{-1}}} \\ &= \frac{4}{3} \frac{z^2 e^2}{m^2 c^3} \sqrt{\frac{m}{2}} (zZe^2)^{3/2} r_{\min}^{1/2} \int_{r_{\min}}^{+\infty} r^{-7/2} (r - r_{\min})^{-1/2} dr \\ &= \frac{4}{3} \frac{z^2 e^2}{m^2 c^3} \sqrt{\frac{m}{2}} (zZe^2)^{3/2} r_{\min}^{1/2} \frac{16}{15r_{\min}^3}, \end{aligned}$$

where we have used the result that

$$\int_a^{+\infty} \frac{x^{-7/2}}{\sqrt{x-a}} dx = \frac{16}{15a^3}.$$

Since

$$E = \frac{1}{2}mv_0^2 = V(r_{\min}) = \frac{zZe^2}{r_{\min}},$$

we have

$$r_{\min} = \frac{2zZe^2}{mv_0^2}.$$

Substitute $r_{\min}$ into the above formula, we will get

$$\Delta W = \frac{64}{45} \frac{z^2 e^2}{m^2 c^3} \sqrt{\frac{m}{2}} (zZe^2)^{3/2} \left(\frac{2zZe^2}{mv_0^2} \right)^{-5/2} = \frac{8}{45} \frac{zmv_0^5}{Zc^3}.$$

The energy loss compared to the total energy is

$$\frac{\Delta W}{E} = \frac{16}{45} \frac{zv_0^3}{Zc^3} \sim \left(\frac{v_0}{c} \right)^3 \ll 1,$$

which justifies the use of energy conservation to obtain the velocity of the particle.